

PAPER_520 — Session 140 Hub: grok_share_0f5d4c91f2c.txt — DPM Shell-Energy Radiance, Negative Time, and DPM-Unified Forces

Unified Quantum Field Framework (UQFF)

Daniel T. Murphy • UQFF Research Consortium • 2026

Source: PAPER_520_Session140_Hub_DPM_Shell_Radiance_Negative_Time_Forces.md

PAPER_520 — Session 140 Hub: grok_share_0f5d4c91f2c.txt — DPM Shell-Energy Radiance, Negative Time, and DPM-Unified Forces

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§1 — Session Overview

Source document: [grok_share_0f5d4c91f2c.txt](#) **Origin:** BigBangHypergraphTheory_12Dec2025.docx follow-up recalculation **Position in pipeline:** Continuation of Session 136 (BigBangHypergraph); recalculation incorporating DPM correction, negative time proof, and force unification.

Papers generated: PAPER_516–520 (5 papers) **CP4 classes introduced:** #111–#115 (5 classes + this hub)

§2 — Corrections to Session 136

Session 136 encoded the [BigBangHypergraphTheory_12Dec2025.docx](#) content (fully integrated into the codebase). Session 140 introduces the following **corrections and refinements** from the Grok recalculation follow-up:

#	Item	Prior Form	Session 140 Upgrade		
1	DPM encapsulation	"SCm encapsulates"	DPM reaction forms layered shell-energies		
2	Phase cascade	Unordered	quantum-multi-f ields→plasma→ gas→liquid→sol id		
3	$t_{\{\text{adj}\}}\$$	$t_{\{\text{obs}\}}/(1+\Delta_{\{\text{rel}\}})\$$	$t_{\{\text{obs}\}}/(1+\Delta_{\{\text{dil}\}}) + t_{\{\text{neg}\}}\$$		
4	Spooky distance	Qualitative only	$\$Distance_{\{\text{spooky}\}} = c \cdot t_{\{\text{neg}\}}\$$	$t_{\{\text{neg}\}}$	$\$$
5	Dual existence	Not defined	$\$DualExist = \ln t_{\{\text{pos}\}}^{\{\text{neg}\}} Existence, dt\$$		
6	$\$F_{\{\text{inert}\}}\$$	Not a pure force	$\$-\partial(DPM_{\{\text{react}\}} \cdot SE)/\partial v^{26} \cdot t_{\{\text{neg}\}}\$$		
7	$\$F_{\{\text{centrip}\}}\$$	$\$m \omega^2 r\$$ (classical)	$\$DPM_n \cdot \omega_{CW}^2 \cdot r^l / (1+\Delta_{\{\text{dil}\}})\$$		
8	$\$F_{\{\text{centrif}\}}\$$	Fictitious (classical)	$\$DPM_s \cdot \omega_{CCW}^2 \cdot r^l \cdot t_{\{\text{neg}\}}\$$ (pure)		

#	Item	Prior Form	Session 140 Upgrade		
9	$\$Prob_{\{\text{or der}\}}\$$	$\$(v_i - v_c)\$$ factor only	$\$times (1 + \Delta t_{\{\text{dil}\}} \cdot t_{\{\text{neg}\}})\$$		

§3 — New Physics Assets (Session 140)

PAPER_516 — DPM Layered Shell-Energy Radiance Phase Cascade

CP4 #111 — DPMLayeredShellEnergyRadianceCalculator

$$ShellEnergy^{\{l\}} = \int Radiance_{\{\text{quant}\}} \cdot dt_{\{\text{neg}\}} \quad \$DPM_{\{\text{react}\}} = \frac{\kappa(DPM_n - DPM_s)}{r^{26}} + \frac{\partial^{26}}{\partial Grind_{\{\text{opp}\}}} \partial t^{26}_{\{\text{adj}\}} \quad \$$$

Triple-calc: Layer 1 (CW), Layer 2 (CCW), Layer 3 ($t_{\{\text{neg}\}}$). Phase cascade: quantum-multi-fields → plasma → gas → liquid → solid.

PAPER_517 — Negative Time Dilation Proof

CP4 #112 — NegativeTimeDilationSpookyDistanceCalculator

$$t_{\{\text{adj}\}} = \frac{t_{\{\text{obs}\}}}{1 + \Delta_{\{\text{dil}\}}} + t_{\{\text{neg}\}} \quad \$Distance_{\{\text{spooky}\}} = c \cdot |t_{\{\text{neg}\}}| \quad \$DualExist = \int t_{\{\text{pos}\}}^{\{t_{\{\text{neg}\}}\}} Existence, dt \quad \$$$

Observable $\Delta_{\{\text{dil}\}} \neq 0$ proves $t_{\{\text{neg}\}} < 0$.

PAPER_518 — DPM-Unified Forces

CP4 #113 — DPMUnifiedInertiaCentripetCentrifugCalculator

$$F_{\{\text{inert}\}} = -\frac{\partial(DPM_{\{\text{react}\}} \cdot ShellEnergy)}{\partial v^{26}} \cdot t_{\{\text{neg}\}} \quad \$F_{\{\text{centrip}\}} = \frac{DPM_n(SCm)}{r^{26}} \cdot \omega_{\{CW\}}^2 \cdot r^{26} \cdot \frac{\partial}{\partial t_{\{\text{neg}\}}} \quad \$F_{\{\text{centrif}\}} = DPM_s(UA') \cdot \omega_{\{CCW\}}^2 \cdot r^{26} \cdot t_{\{\text{neg}\}} \quad \$F_{\{\text{inert}\}} = F_{\{\text{centrip}\}} - F_{\{\text{centrif}\}} \quad [M = F_{\{\text{inert}\}}/a^{26}] \quad \$$$

Resolves classical fictitious-force conundrum: all 3 are pure DPM-emergent.

PAPER_519 — Shell Radiance Prototype Equation

CP4 #114 — ShellRadiancePrototypeEquationCalculator

Full assembled system: ProtoH, \$U_b\$, BigBang trigger, \$Prob_{\text{order}}\$ with \$(1+\Delta_{\text{dil}})\cdot t_{\text{neg}}\$ factor. Master form:

$$\Psi_{26D}(t_{\text{adj}}) = \text{ProtoH} + U_b \cdot Prob_{\text{order}}$$

- BigBang $\cdot \exp\left(-\frac{|t_{\text{neg}}|}{t_{\text{adj}}}\right)$

§4 — CP4 Registry Update

Class	#	Paper	Status
DPMLayeredShellEnergyRadianceCalculator	111	PAPER_516	Implemented
NegativeTimeDilationSpookyDistanceCalculator	112	PAPER_517	Implemented
DPMUnifiedInertiaCentripetCentrifugCalculator	113	PAPER_518	Implemented
ShellRadiancePrototypeEquationCalculator	114	PAPER_519	Implemented
Session140GrokShare0f5d4c91f2cHubCalculator	115	PAPER_520	Implemented

CP4 total classes: 108 (103 prior implementations + 5 Session 140) CP4 all entries: 115 (110 prior + 5 Session 140)

§5 — OutputData Registration

SOURCE180_SESSION140_RESULTS (document_id=25) registered in CondensedPhysics_OutputData.py with complete equation set, 8 new physics items, mass equilibrium, triple-calc system, canonical constants, and 5/5 validation tests passed.

§6 — Integration Confirmation

All Session 140 physics verified **not present** in pre-existing codebase:

- No DualExist math (prior: QuantumEntanglementTerm qualitative only)

- No `Distance_spooky = c · |t_neg|` (prior: qualitative spooky reference only)
- No DPM-based $F_{\text{inert}}/F_{\text{centrip}}/F_{\text{centrif}}$

(prior: classical $m\omega^2 r$ form in `compute_centripetal_centrifugal()`)

- No t_{adj} with t_{neg} term (prior: missing that term)
- No $(1+\Delta_{\text{dil}}) \cdot t_{\text{neg}}$ factor in $\text{Prob}_{\text{order}}$
- No ordered phase cascade (prior: unordered)

Session 140 integration is additive and backward-compatible.

CP4 v5.00 — Session 140 complete.